

Executive Summary & Project Vision

AUTHENTIC INDIAN DATA

Trained on **27,614 verified residential properties** across Mumbai, Bangalore, Delhi-NCR, Hyderabad, Chennai & Kolkata.

PRODUCTION ML PIPELINE

Trained Ridge, Random Forest, GBR & XGBoost with log-target transformation. Best Test MAE: **₹41.89 Lakhs**.

CONFORMAL UNCERTAINTY

Eliminated naive $\pm$ RMSE errors. Built split conformal prediction intervals with **79.75% empirical test coverage**.

■ Core Objectives & Technical Deliverables

- **Eliminate Pricing Asymmetry:** Provide transparent algorithmic price discovery for homebuyers, sellers, and mortgage lenders.
- **Native Indian Rupee (₹) Denomination:** Predict real property values in Lakhs and Crores with ₹/sq.ft micro-market rates.
- **Interactive PropTech Web Platform:** Deploy a full-featured Streamlit UI with single appraisal, market analytics, bulk CSV, and EMI calculator.

Problem Statement: Challenges in Indian Real Estate

■ Traditional Valuation Pitfalls

- **Subjective Broker Quotes:** Manual appraisals rely on word-of-mouth broker inquiries, leading to 15%–30% quote inflation.
- **Extreme Price Skewness:** Indian residential properties range from ■20 Lakhs to ■80+ Crores; linear models break down on heavy right-tail variance.
- **The 'Ames / Foreign Data Fallacy':** Legacy ML tutorials map US variables (snow porches, basements, Iowa zoning) to Indian cities, creating artificial synthetic results.

■ The BharatProp AI Solution

- **Empirical Multi-City Ground Truth:** Trained on 27,614 verified listings across Mumbai, Bangalore, Delhi, Hyderabad, Chennai, and Kolkata.
- **Domain Feature Engineering:** Super built-up area, BHK configurations, 35 gated community amenities, and vastu compliance.
- **Continuous Target Encoding:** Learns micro-market price shrinkage for 1,711 localities without data leakage.
- **Calibrated Uncertainty Bounds:** Delivers defensible, mathematically validated prediction ranges.

Dataset Overview: 6 Metropolitan Economic Hubs

Mumbai	Maharashtra	6,499	■92.00 L	995 sq ft	■10,086/sq.ft
Delhi-NCR	Delhi / NCR	3,851	■75.00 L	1,282 sq ft	■6,945/sq.ft
Bangalore	Karnataka	5,364	■73.61 L	1,492 sq ft	■5,400/sq.ft
Hyderabad	Telangana	1,993	■78.75 L	1,667 sq ft	■5,112/sq.ft
Chennai	Tamil Nadu	4,183	■59.25 L	1,221 sq ft	■5,475/sq.ft
Kolkata	West Bengal	5,724	■51.25 L	1,206 sq ft	■4,555/sq.ft

■ Dataset Specifications & Features

- **Total Verified Properties:** 27,614 | **Distinct Micro-Markets:** 1,711 | **Target:** Price in Indian Rupees (■)
- **35 Binary Amenities:** Gymnasium, Swimming Pool, 24x7 Security, Power Backup, Lift, Piped Gas, Vaastu, Club House, AC, etc.
- **Source:** Kaggle & Open Indian Housing Repository | **License:** Open Access CC0 for research & commercial prototyping.

End-to-End Machine Learning Architecture

1. Data Ingestion

Multi-city CSV download, deduplication & realistic boundary filters (■1.8k–48k/sqft).

2. Feature Engineering

Calculate Area/BHK ratio, BHKxArea interaction, Furnishing, Security & Amenity scores.

3. Preprocessing

ColumnTransformer with OneHot (City), continuous TargetEncoder (Location), RobustScaler.

4. Log-Target Regressor

TransformedTargetRegressor with log1p forward and expm1 inverse transforms.

5. Multi-Model 5-Fold CV

5-Fold Cross Validation comparing Ridge, Random Forest, Gradient Boosting & XGBoost.

6. Conformal Calibration

Empirical non-conformity quantile calculation on calibration fold (N = 4,143 properties).

Domain Feature Engineering & Zero-Leakage Preprocessing

■ Engineered Domain Features

- **Area_per_BHK:** Built-up square footage divided by room count (indicates spaciousness and luxury tier).
- **BHK_x_Area Interaction:** Captures non-linear price scaling between spaciousness and bedroom density.
- **Furnishing_Score (0–9):** Sum of verified home appliances (AC, TV, Refrigerator, Washing Machine, Microwave).
- **Security_Score (0–5):** Composite index of 24x7 security, intercom, power backup, maintenance, and lifts.
- **Recreation_Score (0–9):** Gated community lifestyle index (gym, pool, clubhouse, sports, green gardens).

■ ColumnTransformer & Zero Leakage

- **Micro-Market Target Encoding:** TargetEncoder(target_type='continuous', smooth='auto') fitted strictly on training folds for 1,711 localities.
- **City One-Hot Encoding:** Unbiased binary categorical encoding across all 6 metropolitan hubs.
- **RobustScaler:** Centers features using IQR to prevent multi-Crore outliers from distorting gradients.
- **No Target Leakage:** Derived price metrics (■/sq.ft) are computed strictly after prediction.

Regression Models & Log-Target Optimization

■ XGBoost Regressor

Extreme gradient boosted decision trees with depth pruning (`max_depth=6`), `colsample` subsampling (0.85), learning rate 0.06, and 220 estimators. Best generalizer.

■ Gradient Boosting

Sequential additive decision trees with shrinkage rate 0.08 and 175 estimators. Strong baseline with lowest test RMSE.

■ Random Forest

Bagging ensemble of 150 de-correlated trees (`max_depth=14`) reducing variance across diverse property clusters.

4■■ Ridge Regression

L2-regularized linear model (`alpha=10.0`) establishing linear baseline across encoded location priors.

■ The Log-Target Transformation Principle

- Property prices are heavily right-skewed: fitting models on raw rupees leads squared errors to over-focus on multi-Crore luxury estates.
- Wrapping models in `TransformedTargetRegressor(func=log1p, inverse_func=expm1)` minimizes relative proportional percentage errors across all housing tiers.

Model Benchmarking & Evaluation Leaderboard

■ XGBoost Regressor	■41.89 L	■98.04 L	0.3921	0.4664	0.4788	41.71%
■ Gradient Boosting	■42.02 L	■96.64 L	0.4093	0.4655	0.4786	42.18%
■ Random Forest	■42.29 L	■99.96 L	0.3680	0.4499	0.4643	42.69%
4■■ Ridge Regression	■44.31 L	■102.01 L	0.3419	0.4260	0.4418	44.81%

■ Key Evaluation Takeaways

- **Holdout Test Generalization:** All metrics evaluated on N = 4,143 independent properties (15% split) unseen during training.
- **XGBoost Superiority:** XGBoost achieved the lowest Test MAE (■41.89 Lakhs) and lowest MAPE (41.71%) with consistent 5-fold CV stability (0.4788).
- **Understanding R²:** R² is the coefficient of determination (explaining ~47% of logarithmic variance across 1,711 micro-markets); it is NOT a percentage accuracy.

Prediction Uncertainty: Split Conformal Prediction

■ The Flaw of Naive $\pm$ RMSE

- **Falsely Assumes Constant Normal Errors:** Real estate errors scale with property value (a ₹2 Cr house has wider absolute dollar error than a ₹30 L flat).
- **Produces Negative Valuations:** For affordable properties, subtracting a fixed global RMSE can produce mathematically impossible negative lower bounds.
- **Zero Statistical Guarantee:** $\pm$ RMSE does not guarantee 80% or 95% coverage on real-world test distributions.

■ The Conformal Prediction Method

- **Inductive Split Calibration:** Non-conformity scores $s_i = |\log(y_i) - \log(\hat{y}_i)|$ calculated on $N = 4,143$ validation properties.
- **Exact 80% Empirical Coverage:** Test set empirical coverage on unseen properties is 79.75% (matching 80% nominal target).
- **Multiplicative Prediction Ranges:** Yields intuitive asymmetric ranges $[\hat{y} \cdot \exp(-q), \hat{y} \cdot \exp(+q)]$ preventing negative prices.
- **Honest UI Labeling:** Presented explicitly as a 'Prediction Range / Interval', NOT a confidence interval.

Interactive Web Application (Streamlit)

■ 1. Overview & Methodology

Project background, Indian market dynamics, pipeline architecture diagram, and data summaries.

■ 2. Property Valuation

Real-time appraisal with dynamic city/locality dropdowns, preset selectors, ■/sq.ft rates & 80% conformal range.

■ 3. Model Performance

Leaderboard table, diagnostic plots (Actual vs Pred, Residuals, Bar Charts) & metric explanations.

■ 4. Market Analytics

Interactive multi-city pricing histograms, Area vs Price scatter plots, BHK boxplots, and top locality rankings.

■ 5. Bulk CSV Valuation

Batch portfolio appraisal via CSV upload with schema validation and downloadable result CSVs.

■ 6. Home Loan EMI Planner

Down payment slider, LTV calculation, interest vs principal pie chart, and 10-year amortization schedule.

Real-World Applications & Industry Impact

■■■■■ Homebuyers & Sellers	Instant fair-market benchmark protecting buyers from broker quote inflation and providing negotiation leverage.
■ Banks & Housing Finance (HFCs)	Instant collateral pre-screening for home loan underwriting to verify statutory 75%–80% Loan-to-Value (LTV) limits.
■ PropTech Portals (e.g. 99acres)	Powers automated 'Instant Price Estimate' badges and guides sellers on optimal listing pricing.
■ Real Estate Funds & REITs	Bulk portfolio stress-testing via CSV batch evaluation during quarterly asset audits and acquisitions.
■■ Real Estate Developers	Data-driven pricing for new residential launches by quantifying specific amenity valuation premiums.

Project Limitations & Professional Disclaimer

■ ■ Technical & Market Limitations

- **Asking vs Transaction Price Gap:** Data reflects asking/listing prices; actual negotiated transaction prices may be 5%–15% lower.
- **Unobserved Micro-Location Factors:** Tabular models cannot observe floor-rise premiums, corner plots, ocean views, or builder brand prestige.
- **Market Dynamics & Inflation:** Historical listing data does not automatically factor sudden interest rate or municipal tax changes.

■ ■ Official Professional Disclaimer

STATISTICAL ESTIMATE ONLY:

The valuations and prediction intervals generated by BharatProp AI are automated statistical approximations intended solely for informational, research, and educational purposes.

They do NOT constitute certified legal appraisals under RICS / Government guidelines, title search opinions, banking mortgage approvals, or tax advice. Users should consult licensed property evaluators prior to executing transactions.

Future Innovations & Product Roadmap

■ ■ Geospatial & Satellite Intelligence	Integrate OpenStreetMap and GIS coordinates for distance-to-metro, proximity to IT corridors, and green space ratios.
■ Computer Vision Image Valuation	Train multi-modal CNN/Vision Transformers on floor plans and interior photos to estimate furnishing quality and wear.
■ RERA & Land Registry Integration	Connect with state RERA and Sub-Registrar transaction portals to blend asking prices with actual registered circle rates.
■ RESTful Microservice API	Deploy a FastAPI backend to integrate real-time property valuation endpoints into external fintech and banking apps.

